

Auteur: Mark Gijsbrechts
Studierichting: Informatica
Oefeningenreeks 1B nr. 22.3

Bereken $z + z^{-1}$ voor $z = 2 + 3i$

$$\begin{aligned} &= 2 + 3i + \frac{1}{2 + 3i} \\ &= \frac{(2 + 3i)(2 + 3i) + 1}{2 + 3i} \\ &= \frac{4 + 6i + 6i + 9i^2 + 1}{2 + 3i} \\ &= \frac{4 - 9 + 12i}{2 + 3i} \\ &= \frac{-4 + 12i}{2 + 3i} \\ &= \frac{(-4 + 12i)(2 - 3i)}{4 - 9i^2} \\ &= \frac{-8 + 12i + 24i - 36i^2}{4 + 9} \\ &= \frac{28 + 36i}{13} \end{aligned}$$

References